

Localized Telco Retention: A Geospatial and Financial Risk Analysis

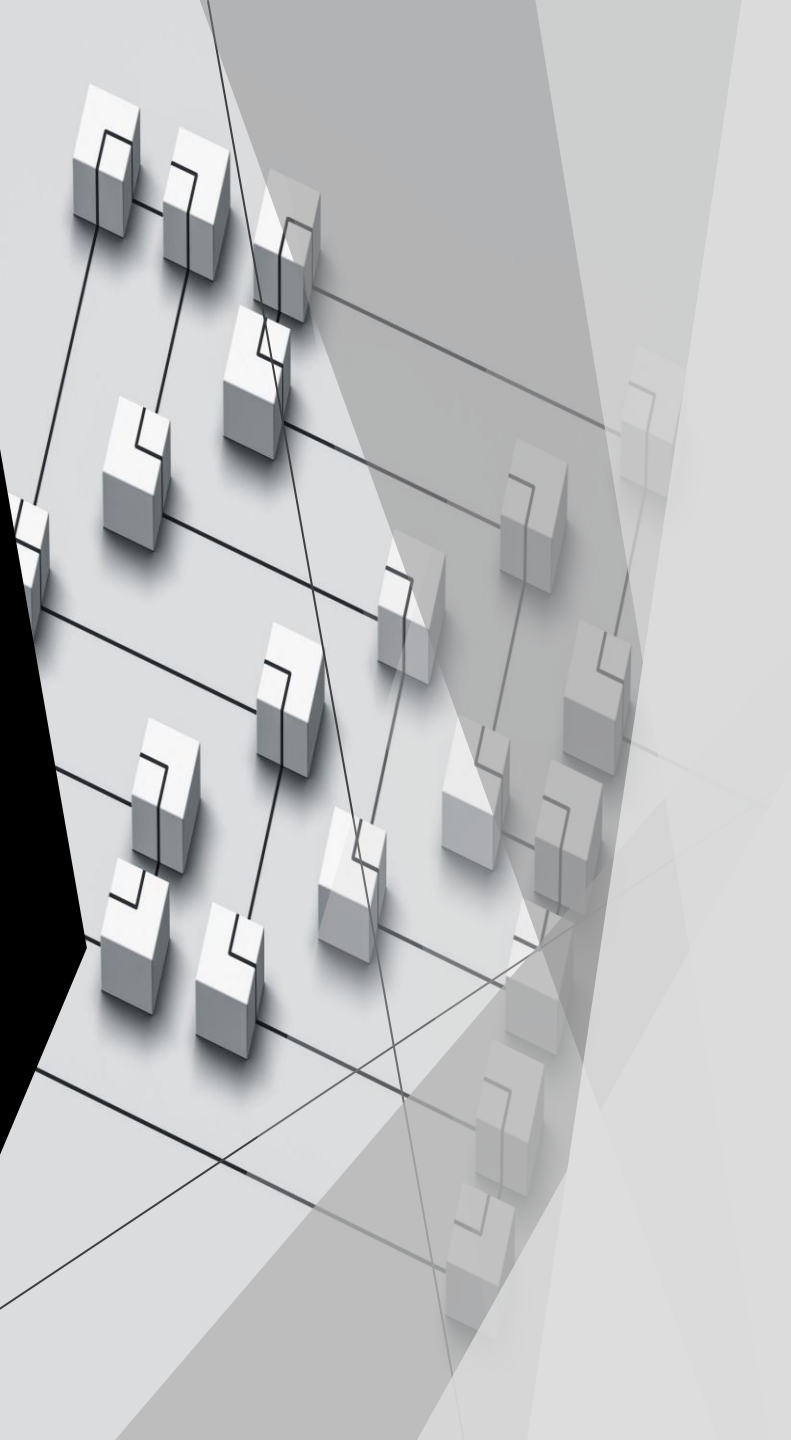
(A Data-Driven Strategy Using Network Quality and Targeted Contracts to Reduce Customer Loss)



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Executive Summary

- ▶ **The Problem:** Traditional churn tracking relies on lagging billing data, ignoring real-world physical network quality that directly causes customer loss.
- ▶ **The Innovation:** A hybrid tracking model combining customer billing details with active local network performance indicators:
 - ▶ Signal Stability Index
 - ▶ Outage History Score
 - ▶ Competitor Exposure
- ▶ **The Strategy:** Tuned a clear, transparent model to prioritize finding at-risk accounts, setting our risk threshold at **0.42** to lock in **85% recall** (capturing 85% of vulnerable subscribers).
- ▶ **Key Discovery:** Fiber optic lines are our highest risk area. While poor signal stability does not directly cause customer loss on its own, it acts as a secondary trigger that pushes frustrated premium subscribers to cancel.

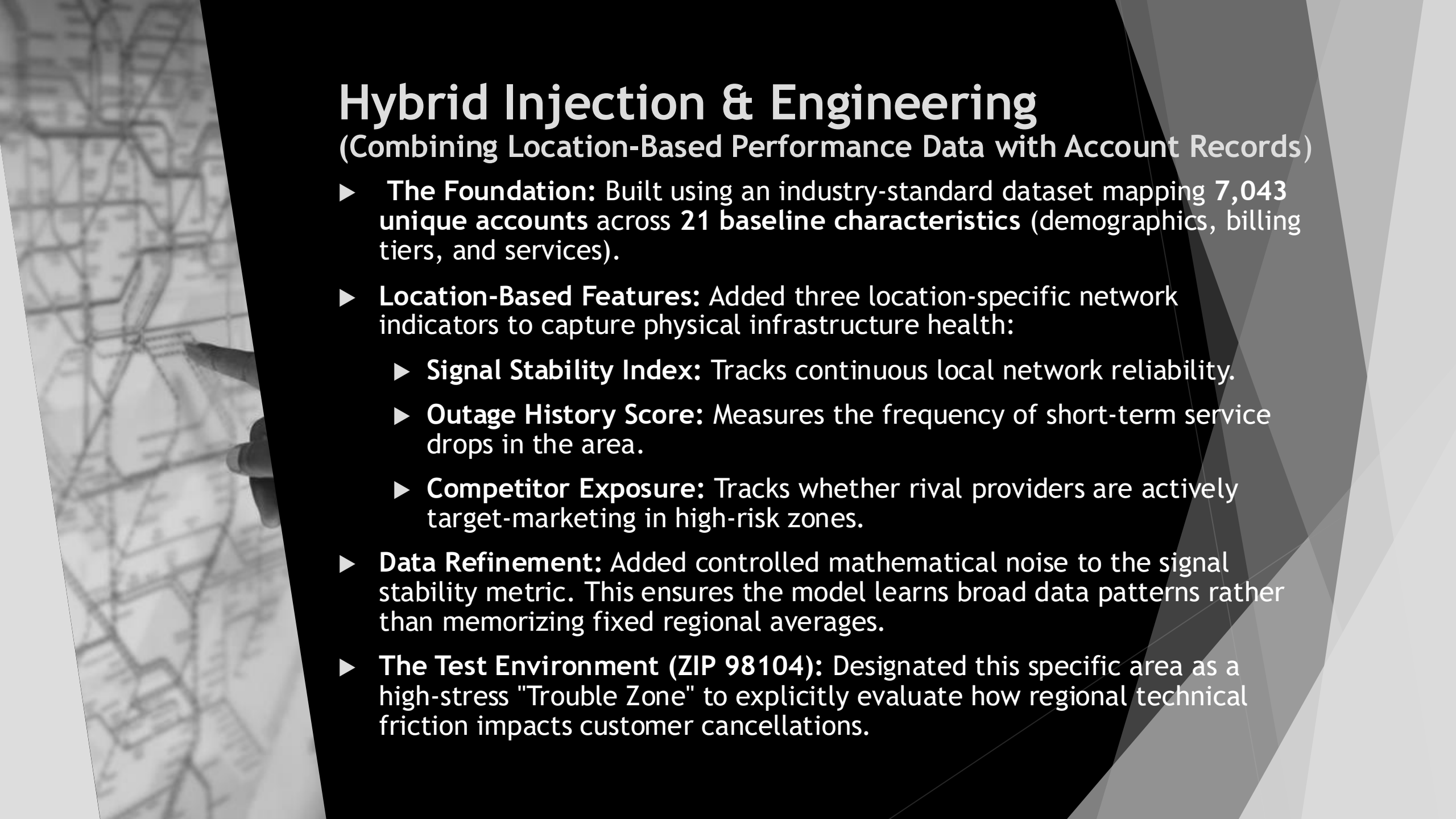




Introduction & Objectives

(Using Network Quality and Financial Data to Eliminate Our Revenue Blind Spots)

- ▶ **The Operational Gap:** Traditional tracking systems treat customer loss as a behavioral event based on past billing history, missing the physical infrastructure problems that trigger cancellations.
- ▶ **The Hybrid Approach:** Combines standard corporate financial data with active network performance metrics (Signal Stability Index and Outage History Score) to understand the technical causes of customer loss.
- ▶ **The Modeling Goal:** Calibrated the model at a **0.4236** custom threshold to achieve our strict business target of **85.0% Recall**, ensuring we identify the maximum volume of high-risk accounts early.



Hybrid Injection & Engineering

(Combining Location-Based Performance Data with Account Records)

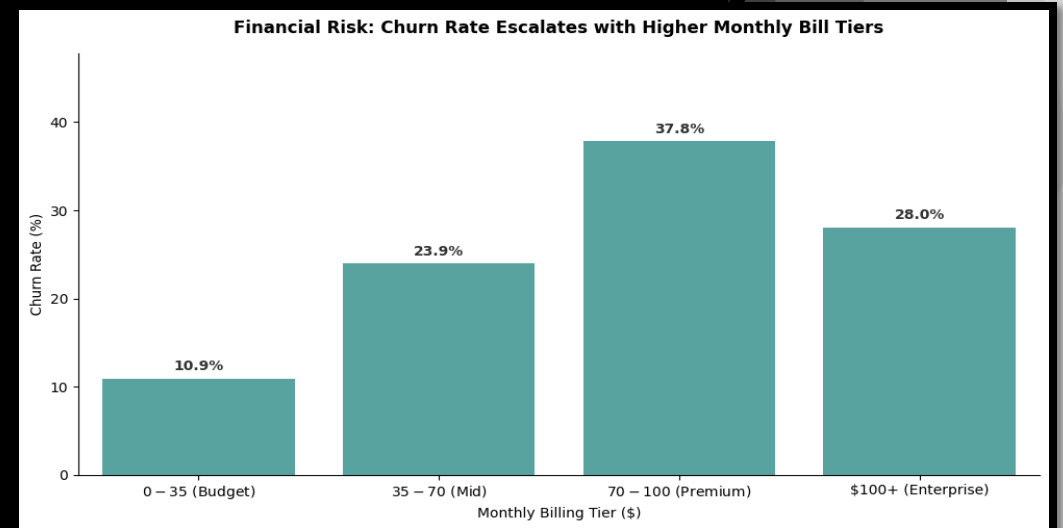
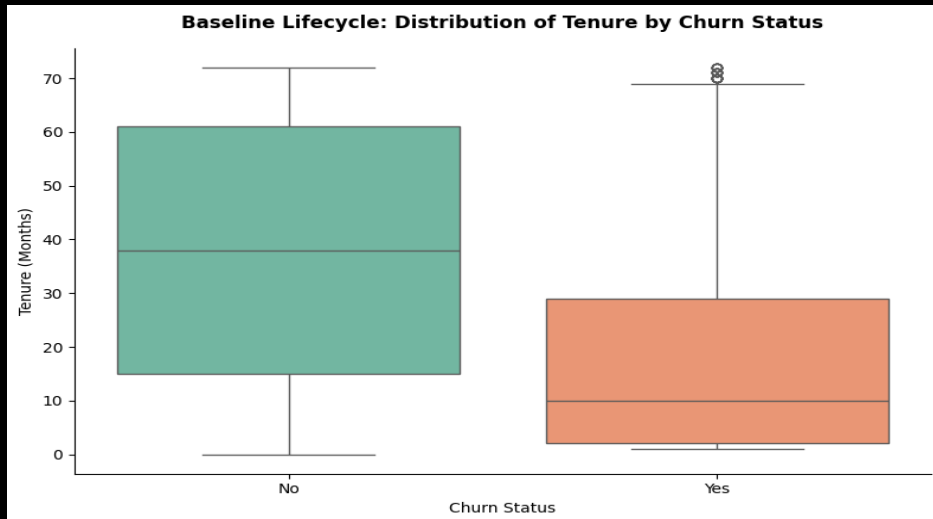
- ▶ **The Foundation:** Built using an industry-standard dataset mapping 7,043 unique accounts across 21 baseline characteristics (demographics, billing tiers, and services).
- ▶ **Location-Based Features:** Added three location-specific network indicators to capture physical infrastructure health:
 - ▶ **Signal Stability Index:** Tracks continuous local network reliability.
 - ▶ **Outage History Score:** Measures the frequency of short-term service drops in the area.
 - ▶ **Competitor Exposure:** Tracks whether rival providers are actively target-marketing in high-risk zones.
- ▶ **Data Refinement:** Added controlled mathematical noise to the signal stability metric. This ensures the model learns broad data patterns rather than memorizing fixed regional averages.
- ▶ **The Test Environment (ZIP 98104):** Designated this specific area as a high-stress "Trouble Zone" to explicitly evaluate how regional technical friction impacts customer cancellations.

Exploratory Data Analysis (EDA)

(The Macro Baseline)

✦ Key Insights:

- ▶ **Early-Stage Vulnerability Window:** 75% of all departing subscribers cancel their service within their first 30 months of activation, identifying a critical onboarding and early-lifecycle risk zone.
- ▶ **The Premium Friction Point:** Risk does not rise at a steady, fixed rate; instead, customer loss peaks severely at 37.8% within the \$70-\$100 premium consumer bracket.
- ▶ **Strategic Indicator:** High-value premium users maintain strict service-quality expectations. If the physical network's quality fails to justify this higher monthly expense, subscribers rapidly cancel their accounts.

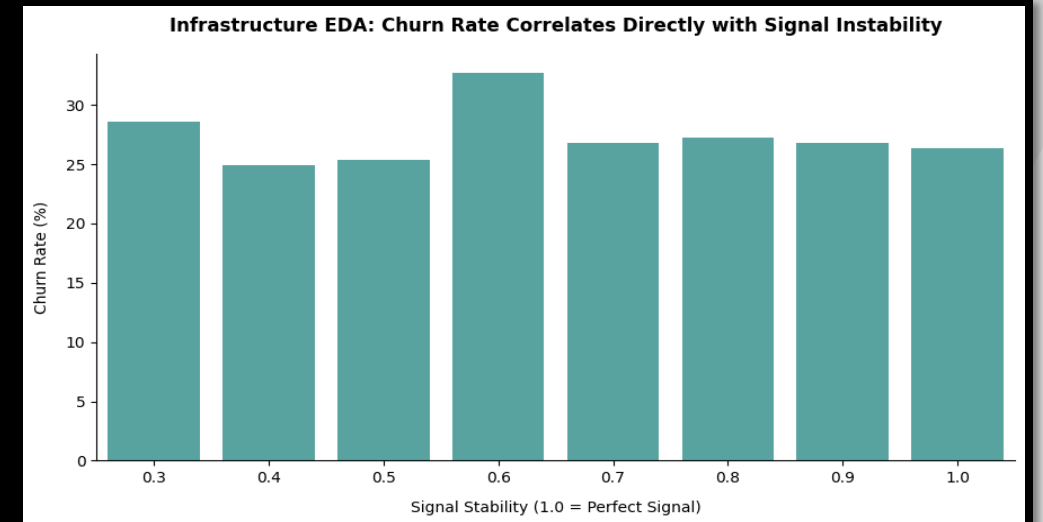
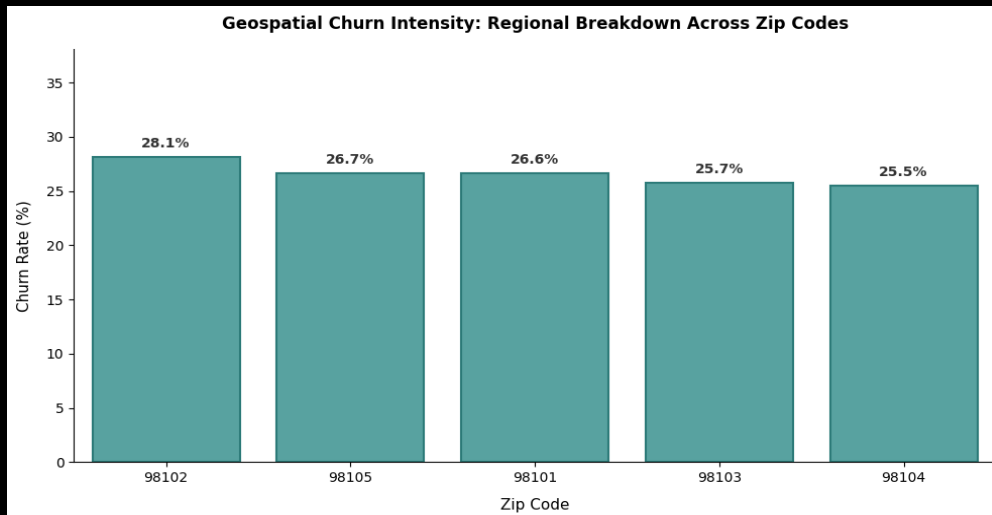


Exploratory Data Analysis (EDA)

(Regional Baselines vs. Technical Signals)

★ Key Insights:

- ▶ **The Regional Baseline:** High-level geography shows a remarkably uniform risk across all five postal zones, hovering tightly between 25.5% and 28.1%. Broad regional boundaries alone cannot tell us why customers leave.
- ▶ **The Telemetry Reality:** Zooming into network performance reveals that customer loss does not rise steadily as signal drops. Instead, risk remains mostly flat (25%-27%) across almost all performance levels, with an isolated spike to ~32% at the 0.6 threshold.
- ▶ **The Contract Safety Net:** Severe signal degradation does not trigger an immediate cancellation. This proves that network drops alone do not drive churn; instead, long-term contracts act as a legal cushion that holds frustrated customers in place temporarily.

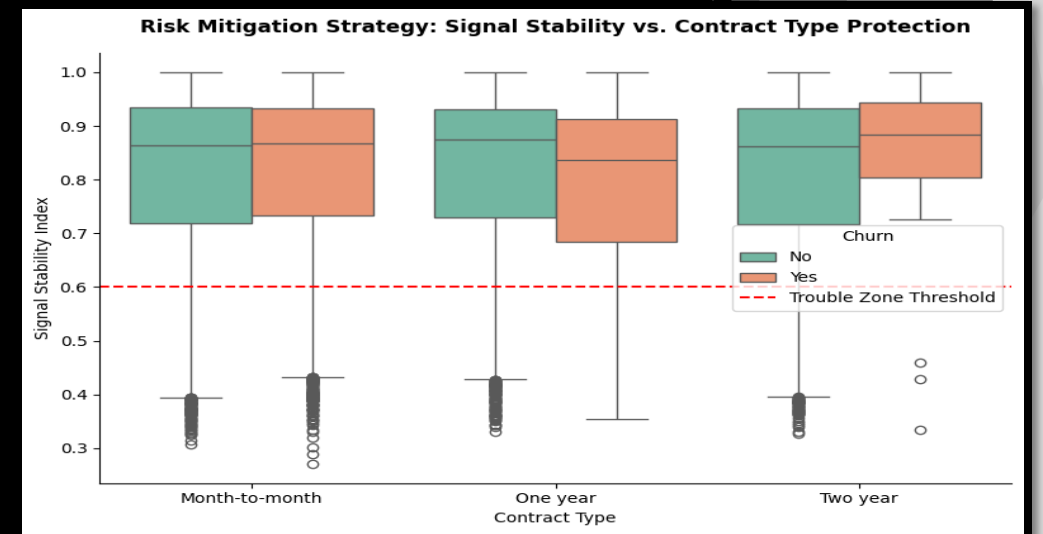
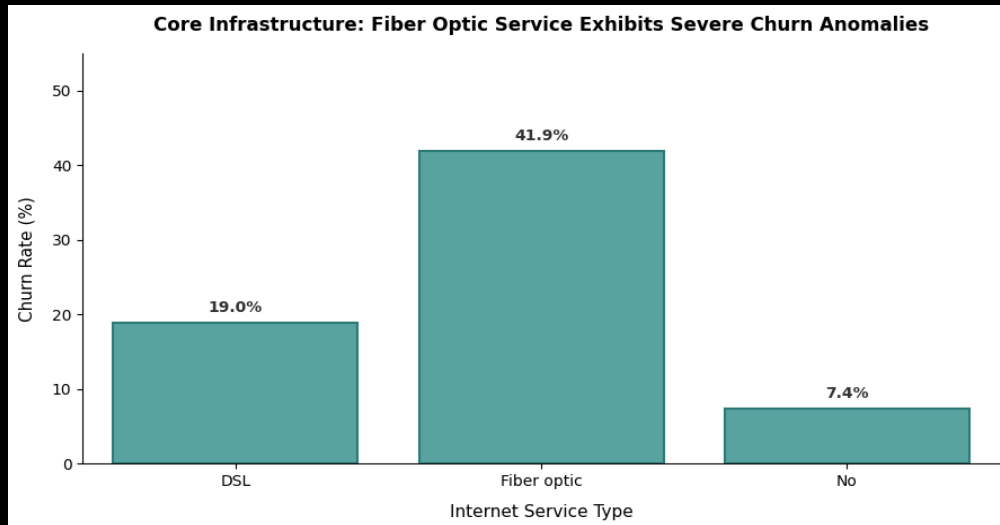


Exploratory Data Analysis (EDA)

(Product Ecosystem & Account Constraints)

📌 Key Insights:

- ▶ **The Fiber Optic Paradox:** Premium fiber plans require our highest capital investment to build but suffer an alarming 41.9% customer loss rate—more than double that of legacy copper DSL (19.0%).
- ▶ **The Legal Protection Net:** Two-year contract agreements act as a highly effective operational cushion. These long-term loyalty commitments temporarily anchor subscribers to our network, even when they experience severe signal drops.
- ▶ **The Hidden Risk:** While long-term contracts secure immediate revenue, they can mask underlying network frustration. If these technical signal issues are not proactively fixed before the terms expire, these premium segments represent a high risk for future cancellation.

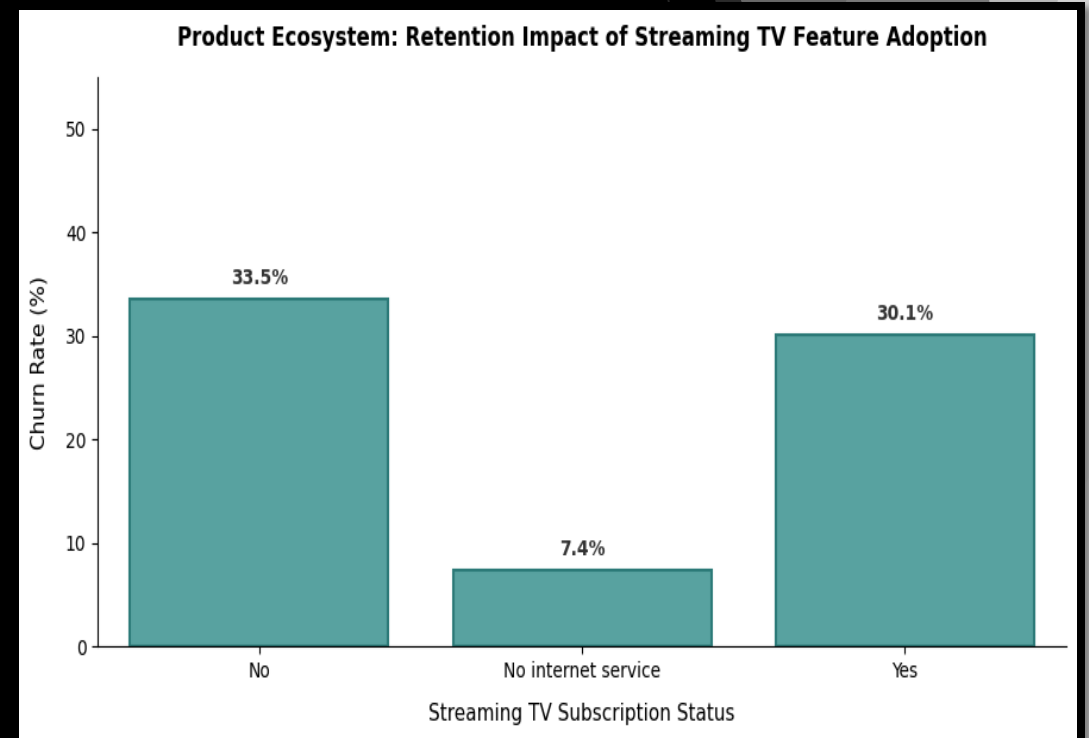


Exploratory Data Analysis (EDA)

(Product Ecosystem - Downstream Value-Added Asset Performance)

📌 Key Insights:

- ▶ **Minor Retention Impact:** Internet subscribers who do not use Streaming TV have a **33.5%** churn rate, while those who add the service drop slightly to **30.1%**.
- ▶ **Diminishing Returns:** Adding software features onto an existing account only provides a minor **3.4 percentage-point lift** in customer retention.
- ▶ **Strategic Takeaway:** Selling digital bundles to at-risk accounts yields a low return on investment if the underlying network connection remains unstable. Corporate resources are better spent stabilizing the physical infrastructure.



Preprocessing & Model Tournament

(Evaluating Options & Finding Our Champion Architecture)

📌 Key Insights:

- ▶ **Leakage Prevention:** Completely separated all customer outcome identifiers from our tracking variables before running any math. This ensures our model mimics a true, unbiased business environment.
- ▶ **Standardizing Data Columns:** Processed all customer categories into numerical indicators, creating a clean, uniform matrix of **36 distinct indicators** to evaluate behaviors.
- ▶ **The Strategic Trade-Off:** While the Gradient Boosting model scored a tiny fraction higher (0.8461 AUC), we deliberately chose **Logistic Regression** (0.8449 AUC) as our champion.
- ▶ **Why Transparency Matters:** A 0.12% accuracy gain is not worth deploying an uninterpretable "black-box" system. Logistic Regression provides completely transparent weights that allow our engineering, supply chain, and marketing teams to trace risk directly to specific infrastructure issues or contracts.

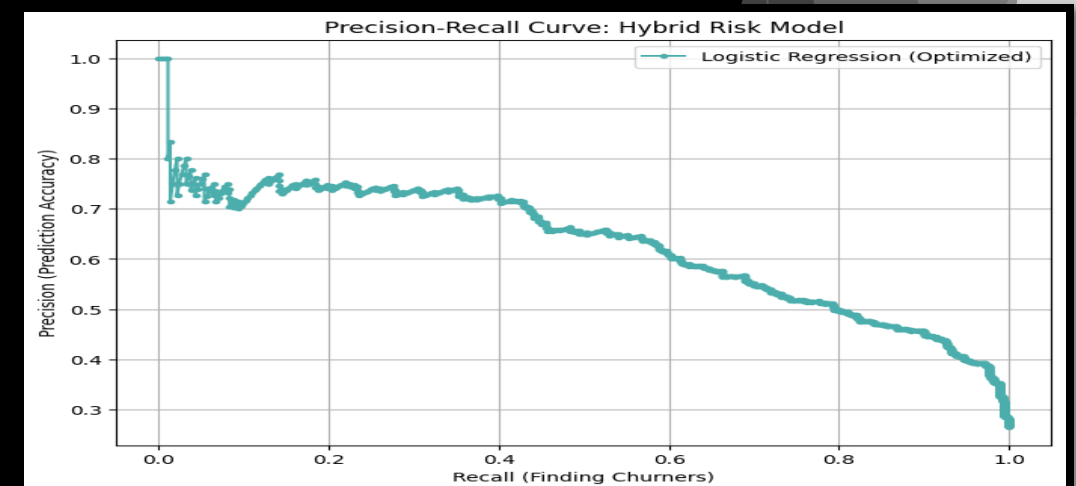
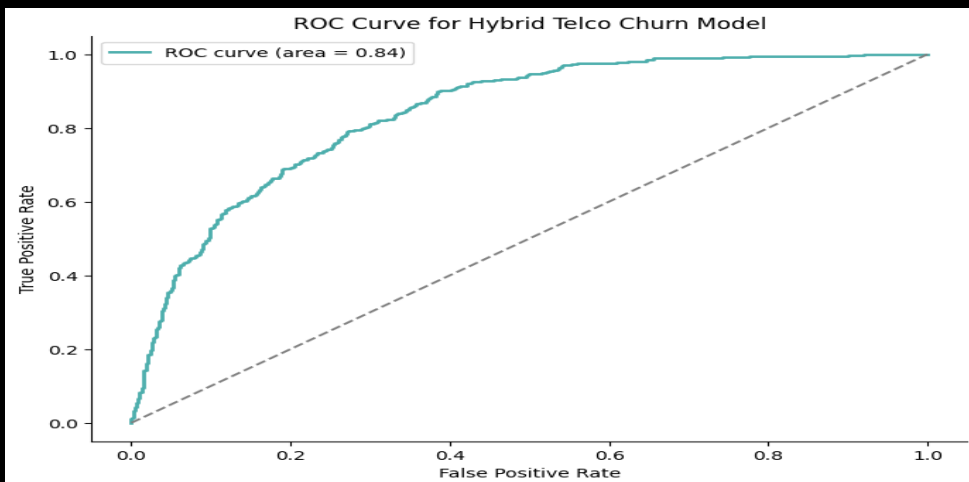
Model	Hyperparameters	Mean ROC-AUC (CV)	Strategic Rationale and Status
Logistic Regression	penalty: elasticnet, C:10, L1_ratio : 1.0	0.8449	Selected Champion: Offers the best balance of clear business transparency and strong predictive power.
Gradient Boosting	n_estimators: 100, learning_rate: 0.1	0.8461	Baseline Benchmark: The highest raw score but maintained for architectural risk comparison.
Random Forest	n_estimators: 200, max_depth: 10	0.8295	Lowest raw score but used for comparison.

Model Performance & Decision Threshold

(Setting the 0.42 Risk Threshold to Catch 85% of At-Risk Account)

✚ Key Insights:

- ▶ **Strong Predictive Accuracy:** The model achieved a robust **0.8410 Area Under the Curve (AUC)** score. This high metric proves the model has strong predictive reliability, successfully separating high-risk customer profiles from stable baseline subscribers.
- ▶ **Calibrating the Risk Threshold:** We intentionally moved our decision threshold down to a **0.42 probability score** to optimize our safety coverage and capture vulnerable accounts early.
- ▶ **The Defensive Business Trade-Off:** This aggressive boundary holds our model's precision at 47.0% to guarantee that we successfully catch 85.0% of all at-risk accounts (Recall).
- ▶ **Cost-Effective Security:** While a lower threshold increases false alarms, the minor operational expense of a proactive retention check is vastly lower than the steep cost of losing premium customer revenue.

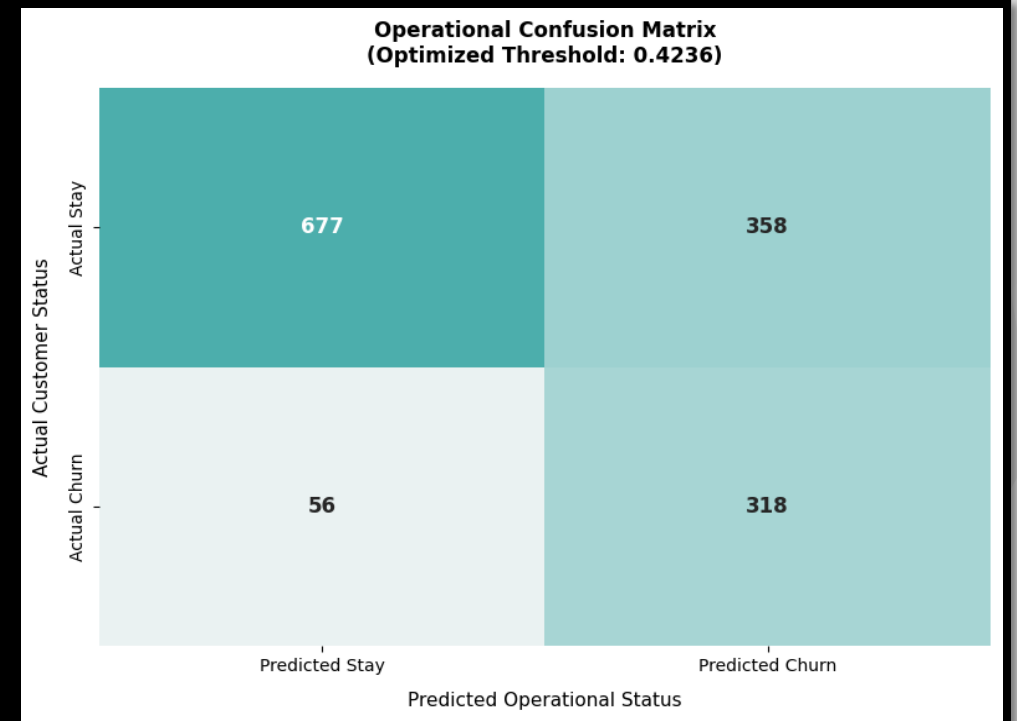


Operational Impact

(Confusion Matrix Volume Analysis)

✦ Key Insights:

- ▶ **318 Confirmed Saves (True Positives):** The model successfully identifies and flags 318 high-value subscribers before they leave, allowing customer success teams to deploy targeted retention offers early.
- ▶ **Tight 15% Miss Rate (False Negatives):** The system completely overlooks only 56 accounts out of the entire churning group. Keeping our blind spot down to a minor 15% rate minimizes sudden, unpredicted revenue losses.
- ▶ **358 False Alarms (False Positives):** The model flags 358 stable subscribers as risk profiles. While this introduces a small amount of extra outreach, sending a low-cost contract renewal incentive or automated signal check to a stable line is vastly cheaper than absorbing account replacement costs.
- ▶ **Protects the Core Business:** This structure ensures our retention resources are concentrated heavily on the highest-risk revenue brackets, moving the company away from costly, blanket discounting strategies.

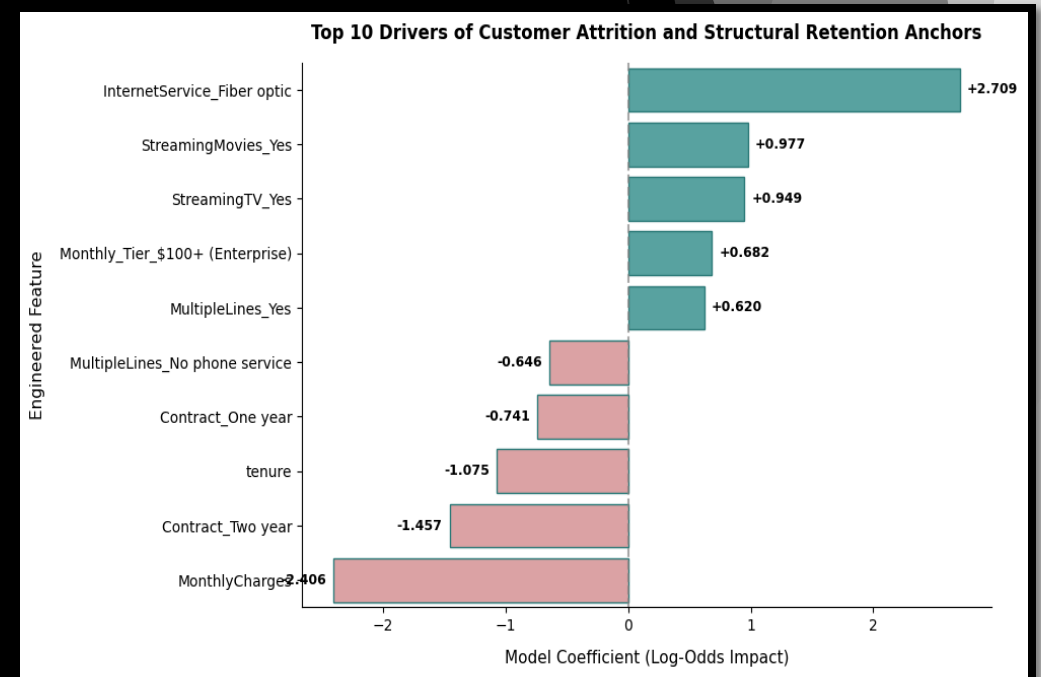


Key Risk Drivers & Retention Anchor

(Understanding What Triggers Cancellations vs. What Locks in Loyalty)

📌 Key Insights:

- ▶ **The Premium Infrastructure Friction Core:** Having a premium **Fiber Optic** plan is the single strongest predictor of customer cancellation across the entire model (+2.709 value weight). High-paying subscribers react strongly to any performance friction.
- ▶ **The High-Value App Segment:** Heavy usage of digital products like streaming movies (+0.977) and streaming TV (+0.949) heavily accelerates customer risk when paired with premium billing tiers exceeding \$100 per month (+0.682).
- ▶ **Our Strongest Defense Lines:** Long-term contractual commitments act as the business's most powerful defensive retention tool. A **Two-Year Contract** (-1.457) and long cumulative customer **tenure** (-1.075) strongly secure revenue and keep subscribers in place.
- ▶ **The Contract Masking Effect:** High continuous monthly charges display a large negative value weight (-2.406). This reveals an interaction effect: when high-spending accounts are secured by two-year agreements, their high spend actually acts as a loyalty anchor rather than a risk factor.

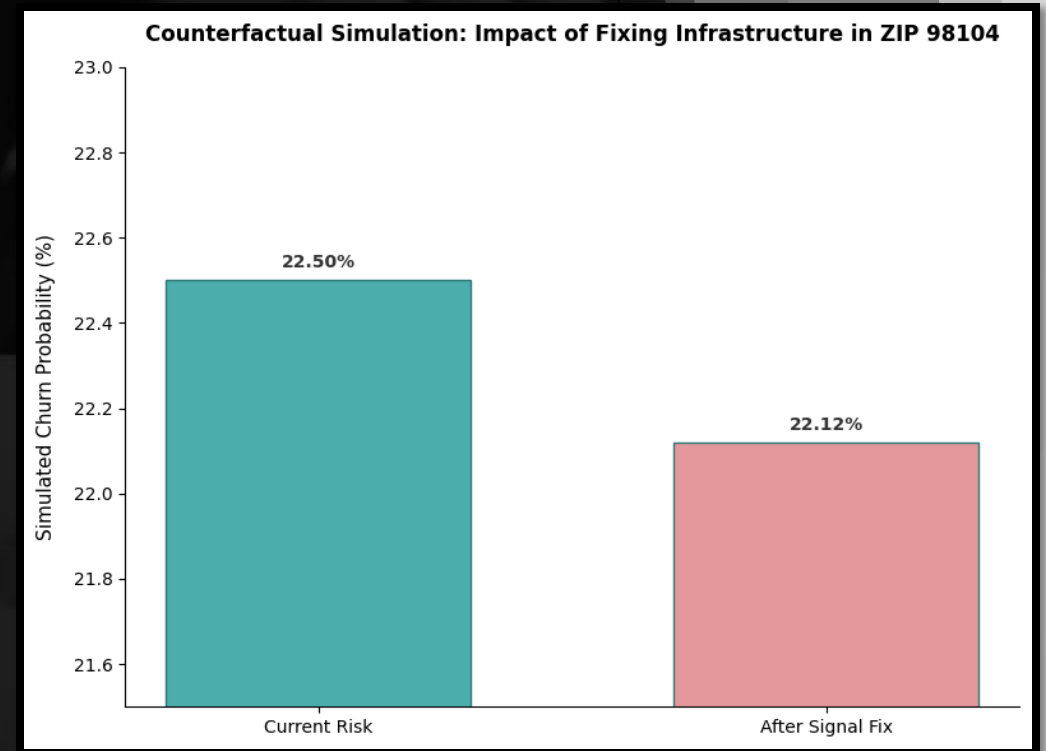


Scenario Simulation & Financial ROI

(Evaluating Capital Expenditures to Stabilize Local Customer Risk)

✦ Key Insights:

- ▶ **The "What-If" Scenario:** Simulated a localized field repair strategy inside our highest-risk network corridor: **ZIP Code 98104**. The model tested the impact of raising the local signal stability index from a failing 0.45 up to an optimal 0.80.
- ▶ **The Modest Financial Return:** Improving the network signal drops the regional customer risk from **22.50% down to 22.12%**. This yields a positive but very modest **1.69% relative risk reduction**.
- ▶ **Confirming the Secondary Trigger:** This modest 1.69% reduction confirms our data analysis: **signal quality alone is not the primary cause of customer cancellation**. It acts as a secondary trigger rather than a standalone driver.
- ▶ **The Structural Connection:** Why prioritize signal fixes if fiber optic plans are our top risk? **Because fiber subscribers in low-signal zones represent the highest-risk intersection in our model**. Stabilizing the signal in these fiber-heavy trouble zones directly defuses our most dangerous revenue risk vector.



Strategic Roadmap & Conclusion

(Shifting from Costly Discounting to Target Network Actions)

- ▶ **Targeted Network Upgrades (CapEx Allocation):** Deploy prioritized engineering audits and fiber hardware upgrades directly inside regional trouble spots, starting with **ZIP Code 98104**. Because fiber subscribers in low-signal zones represent the highest-risk intersection in our model, stabilizing the connection here directly defuses our most dangerous customer risk vector.
- ▶ **Proactive Contract Retention (Revenue Insurance):** Proactively target high-risk, month-to-month subscribers who are paying premium rates. Offer minor loyalty incentives to secure them into formal **Two-Year Contracts**, building a legal buffer to protect immediate revenue while engineering upgrades are completed on the ground.
- ▶ **The Strategic Shift:** Reorients the company away from generic, margin-compressing discount campaigns and shifts toward an intelligent, location-based retention plan. This dual strategy balances physical network fixes with smart customer agreements to protect recurring revenue streams.

Target High-Risk Intersections: Prioritize fiber-heavy trouble zones using our engineered indicators as an automated early-warning system before subscribers cancel.

Stop Margin Compression: Protect high-value revenue corridors through proactive, localized operations instead of relying on expensive, company-wide discounts.

Implement a Dual Strategy: Secure vulnerable month-to-month lines with proactive contract incentives to shield revenue while engineering crews stabilize physical network performance.

Move Beyond Prediction: Shifting the company from simply tracking risk metrics to actively managing and resolving network and product friction at the root.

“Data-driven infrastructure leads to customer-driven loyalty.”

Thank you!

